

Mechanistic simulation of heritable expression states, cell division and drug response in single-cell populations

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Abstract

Heritable, non-genetic differences in gene expression decide which cancer cells survive a drug, yet the simulators used to benchmark single-cell methods model neither cell growth nor division, and therefore cannot generate such states from first principles. Here we present Biomodelling.jl 2.0, a framework in which stochastic reaction kinetics run inside growing, dividing and drug-treated cells. Exact, hybrid and adaptive kernels are coupled to exponential growth, volume-scaled transcription, gene replication and division with binomial partitioning and promoter inheritance, to dose schedules and state-dependent killing, to a complete lineage record, to single-cell observation models and to likelihood-free inference. Expression memory, cell-size scaling and cell-cycle-dependent bursting emerge from this physiology rather than being imposed. A single resistance gene with slow promoter switching reproduces the reported signatures of drug-tolerant persisters, and the framework supplies the ground truth that benchmarks of memory-gene detection, network inference and perturbation prediction lack. Calibrated to time-lapse measurements of cisplatin-treated cells, the model predicts held-out concentrations, lineage correlations and single-cell timing. Used with a built-in optimiser, it reproduces opposite outcomes of intermittent dosing in a melanoma trial and in xenografts, attributes the difference to the fitness of resistant cells without drug, and searches for better schedules.

Isogenic cells differ in their transcriptional states, those differences are often inherited across several divisions, and the inherited state decides what a cell does under stress. Cancer cells that survive targeted or cytotoxic therapy frequently do so without a resistance mutation: they occupy a rare, reversible, drug-tolerant expression state that pre-exists the treatment or is induced by it^{1–5}. Lineage-resolved experiments now measure this memory directly. The fates of sisters and cousins are correlated, clonal barcode diversity is preserved or collapsed depending on whether tolerance pre-exists or is induced, and the timescale of expression fluctuations relative to the cell cycle decides which genes carry memory at all^{6–9}. In parallel, mechanistic analyses of single-cell RNA-sequencing data have shown that transcriptional kinetics can be inferred correctly only when gene copy number, cell age and partitioning at division are modelled explicitly^{10–12}, and evaluations of “virtual cell” models for perturbation prediction keep finding that deep-learning approaches fail to beat simple baselines, largely for want of mechanism and causal structure^{13–16}.

These developments create a demand that existing simulators do not meet. The simulators used to benchmark single-cell methods, among them SERGIO, dyngen, scMultiSim and GRouNdGAN^{17–20}, model regulation and technical noise but not cell growth, volume-dependent transcription, gene replication or the partitioning of molecules at division; lineage simulators such as TedSim²¹ generate division trees but attach phenomenological, largely Markovian expression to them. None of them can therefore produce expression memory, cell-size scaling, cell-cycle-dependent bursting or persister dynamics from first principles, and none provides the ground truth that

the methods now being developed for exactly those phenomena require²². Stochastic simulators of growing and dividing cells exist^{23,24} but stop at the single-cell state: they offer no drug response, no lineage statistics, no observation models, no inference and no benchmark generation.

Here we present Biomodelling.jl 2.0, a rewrite of a Julia framework whose distinguishing feature, the coupling of stochastic reaction kinetics to exponential growth, volume-scaled transcription and binomial partitioning at division²⁵, is precisely the mechanism behind heritable expression states. Version 2.0 turns that feature into a complete framework for the mechanistic study of drug tolerance and for single-cell benchmarking. We validate the engine against exact stationary laws and an independent implementation, show that growth, division and partitioning alone reproduce published results on cell-size scaling, partitioning noise, lineage-versus-population statistics and cell-cycle-dependent bursting, and demonstrate that a single resistance gene with slow promoter switching reproduces the quantitative signatures of drug-tolerant persisters. We then use the framework as a ground-truth generator for memory-gene detection, network inference, imputation and perturbation prediction; as an inference engine that recovers kinetic and drug parameters from snapshot and lineage data and quantifies the bias of ignoring division; as a model calibrated to published time-lapse measurements and validated on conditions it was not fitted to; and, with a built-in schedule optimiser, as a testbed in which clinical dosing regimens are compared and better ones are sought.

Results

A layered simulator of growing, dividing, treated cells

Biomodelling.jl 2.0 is organised in layers (Fig. 1). A model is a list of reactions with mass-action, Hill or custom kinetics and named parameters; promoter states are explicit species, grouped so that they are inherited at division and doubled at replication. Four kernels advance a cell’s state at fixed volume: the exact direct method, fixed-step tau-leaping, a hybrid scheme that falls back to the exact method whenever a leap would produce a negative count, and adaptive tau-leaping with critical-reaction handling. The population layer grows each cell’s volume exponentially, scales propensities with volume so that concentrations are invariant, replicates genes at a set point of the cycle, divides cells by a sizer, adder or timer rule with binomial or over-dispersed partitioning, and controls the population size by Moran-like replacement, free growth or a logistic hazard. A perturbation layer applies dose schedules, state-dependent death, growth inhibition, drug-induced rate changes and genetic perturbations. Every cell’s ancestry, fate and state at birth and division enter a lineage table, from which heritability, autocorrelation, memory timescales, fluctuation-test and clonal-memory statistics follow. Observation models convert true counts into scRNA-seq, smFISH or time-lapse measurements, and the same forward model drives likelihood-free inference. Table 1 sets these capabilities against those of the simulators most used to benchmark single-cell methods.

Table 1. Capabilities of single-cell simulators (✓ native; ○ partial or phenomenological; – absent). ABM.jl, AgentBasedModeling.jl; v1.x and v2.0, Biomodelling.jl versions.

Capability	SERGIO	dyngen	scMulti-Sim	GRouNd-GAN	TedSim	Lineage-Sim	ABM.jl	v1.x	v2.0
Mechanistic reaction kinetics (SSA / tau-leaping)	○	✓	○	–	–	–	✓	✓	✓
Gene regulatory networks with known ground truth	✓	✓	✓	✓	–	–	○	✓	✓
Cell growth, volume-scaled transcription	–	–	–	–	–	–	✓	✓	✓

Capability	SERGIO	dyngen	scMulti-Sim	GRouNd-GAN	TedSim	Lineage-Sim	ABM.jl	v1.x	v2.0
Gene replication / cell- cycle copy number	–	–	o	–	–	–	–	–	✓
Division with molecular partitioning	–	–	–	–	o	o	✓	✓	✓
Heritable promoter states, expression memory	–	–	–	–	o	o	o	o	✓
Lineage tree with per-cell fates	–	–	–	–	✓	✓	o	–	✓
Drug response: dose schedules, state-dependent death	–	–	–	–	–	–	–	–	✓
Genetic perturbations (knockdown / overexpression)	✓	✓	o	✓	–	–	o	–	✓
scRNA-seq observation model	✓	✓	✓	✓	✓	✓	–	o	✓
smFISH / time-lapse observation	–	–	–	–	–	–	–	–	✓
Parameter inference from data	–	–	–	–	–	–	–	–	✓
AnnData output	✓	✓	✓	✓	o	✓	–	–	✓

The engine is exact and reproducible in another package

We first verified the kernels against exact stationary laws (Fig. 2). For a birth-death process the simulated distribution of 50,000 cells is indistinguishable from the Poisson law (Fig. 2a); for the telegraph model in an intermediate and in a bursty regime it matches the Beta-Poisson distribution²⁶ (Fig. 2b); and for bursty protein production it matches the negative binomial (Fig. 2c). Across kernels, Kolmogorov-Smirnov distances to the exact law with 20,000 cells lie below the 99% critical value for the direct method and for the hybrid and adaptive schemes at their default tolerances, at wall-clock costs of 0.25 s to 0.91 s per 10^4 cells (Fig. 2d). Simulating the same telegraph model with JumpProcesses.jl gives distributions whose two-sample Kolmogorov-Smirnov statistic against Biomodelling.jl is 0.0034 (99% critical value 0.0163; Fig. 2e), with a 6.2-fold (in favour of JumpProcesses.jl) difference in serial wall-clock time. Population simulations scale linearly with the number of cells and approximately linearly with the number of reactions: 2.6 s for 5000 cells $\times$ 10 telegraph genes $\times$ 200 steps (hybrid kernel) with four threads (Fig. 2f).

Physiology alone generates memory, size scaling and cell-cycle effects

The physiology of the population layer suffices to produce properties of single-cell data that phenomenological simulators must impose by hand (Fig. 3). In a single lineage of a slowly switching gene, mRNA and protein track volume within a cycle and are halved at division, while the promoter state, inherited by both daughters, persists over many divisions (Fig. 3a). For a fast-switching gene without replication, transcript numbers scale linearly with cell volume, so that concentration is homeostatic; adding mid-cycle gene replication doubles transcription in the second half of the cycle and lifts counts of large cells above the volume-scaling line (Fig. 3b), the copy-number effect that mechanistic fits to scRNA-seq data must account for¹⁰. Correlations of protein concentration at division between mothers and daughters, sisters and cousins fall from 0.94 to 0.02 as the promoter switching rate crosses the inverse cell-cycle time (Fig. 3c), with sisters more correlated than mother-daughter pairs and cousins less, and mRNA less heritable than protein. The memory timescale extracted from the generation-resolved lineage autocorrelation is set by the promoter switching time and is nearly independent of the cell-cycle time as long as memory outlasts the cycle; when the two are comparable the estimate is bounded below by one cell cycle, the resolution of the autocorrelation (Fig. 3d). As reported for real lineages¹¹, the coefficient of variation of a stable protein's concentration is larger across the population than along single lineages, whether or not cell-cycle times

are noisy, and over-dispersed partitioning adds variability in both perspectives (Fig. 3e). Telegraph parameters fitted separately to cells before and after replication differ by the expected factor of two in burst frequency, and a fit that ignores the cell cycle returns intermediate values (Fig. 3f).

One resistance gene reproduces persister phenomenology

To test whether heritable expression states suffice to reproduce the phenomenology of drug-tolerant persisters, we coupled a resistance gene with slow promoter switching (memory timescale 45 time units, cell-cycle time 20) to a death hazard that rises with dose and is suppressed by the concentration of the gene’s protein product (Methods). Under continuous, pulsed and reduced-dose schedules the population first collapses as low-expressing cells die, then regrows from high-expressing survivors (Fig. 4a); kill curves are biphasic with a surviving fraction that falls with dose (Fig. 4b). Between doses 0.5 and 2 the net rate of population decline over the first 30 time units of treatment rises from 0.023 to 0.050 per time unit (2.1-fold), while the mean division time of cells born under drug changes by 0.0% and the mean time to death of killed cells moves only from 8.5 to 6.2 time units (Fig. 4c). This reproduces the observation that population decay depends strongly on dose even though single-cell division and death times barely do⁶, because fate is decided by states that exist before the drug arrives.

Related cells accordingly share the fate of their lineages. Among cells born in the 25 time units before the drug, one whose sister founded a surviving lineage does so itself with probability 0.39, against a marginal probability of 0.08 (4.8-fold enrichment over 1830 sister pairs; 2.9-fold for cousins), whereas in a fast-switching control, in which survival is rarer, the conditional and marginal probabilities coincide (0.03 versus 0.02; Fig. 4d). When tolerance pre-exists, survivors descend from a minority of founder clones and the effective number of clones drops from 491 to 89; when tolerance is drug-induced, so that the drug raises the promoter activation rate of every cell, barcode diversity is largely preserved (165 of 492; Fig. 4e). This is the signature that distinguishes the two origins in barcoding experiments^{2,6}.

Disrupting expression memory changes the outcome, and when it is disrupted matters. Accelerating promoter switching 20-fold from two cell-cycle times before the drug, as memory-disrupting epigenetic compounds do⁹, leaves the number of surviving clones unchanged if the acceleration stops when the drug arrives (98 versus 120 clones), because it reshuffles which cells occupy the resistant state without changing how many do. Maintaining it during exposure reduces the count to 7 clones (10 cells, against 792; Fig. 4f), because resistant cells now leave the protected state within a fraction of the time they need to found a colony. In the model, memory must be disrupted while the drug is present. A transient pulse enriches high-expressing cells and, with slow switching, the enrichment persists for many generations after the drug is removed, whereas with fast switching it relaxes within a few cycles (Fig. 4g): phenotypic selection with stable inheritance, as in the original MGMT model²⁷. Finally, scanning release periods and doses of intermittent schedules (Fig. 4h), the most effective schedule is release period 0, dose 2 (growth rate +0.004 per time unit) for pre-existing tolerance, release period 10, dose 2 (growth rate -0.016 per time unit) when the resistant state carries a fitness cost, and release period 10, dose 1 (growth rate +0.009 per time unit) for drug-induced tolerance. Without a fitness cost nothing beats continuous dosing at the highest dose, since every release period lets the resistant subpopulation regrow. With a fitness cost the population declines under every schedule at dose ≥ 0.5 , and release periods of up to 10 time units match continuous dosing (-0.013 per time unit under continuous dose 2), because cells that leave the resistant state during a holiday are killed on re-exposure. For drug-induced tolerance the ranking of doses reverses: the highest dose induces tolerance in more cells than it kills, so dose 1 outperforms dose 2 at every release period (release period 10, dose 1 (growth rate +0.009 per time unit), against +0.012 for continuous dose 2), the intermediate-dose optimum reported previously²⁸; the benefit of release periods reported in that work does not appear here, because

140 in this model the induced state decays at the same rate with and without drug, so a holiday only postpones the
141 death of reverting cells while all cells grow.

142 **Ground truth for memory genes, networks and perturbations**

143 Because every cell's clone, lineage and true state are known, the framework supplies ground truth that observational
144 data cannot (Fig. 5). For 40 independent genes whose memory times span three decades, MemorySeq-style clonal
145 variance scores computed on true concentrations rise from 1.0 for genes with memory far below the cell-cycle
146 time to 29.6 for the slowest genes, and the ranking survives computation on sequenced counts at 20% capture
147 efficiency (Fig. 5a); the score separates genes with memory longer than the cycle from the rest with an area under
148 the receiver operating curve of 1.00 on true concentrations and 0.99 on sequenced counts. For a 30-gene network
149 simulated at fixed volume, in a dividing population and after sequencing, correlation-based and GENIE3 inference
150 lose most of their accuracy in the dividing population when raw counts are used (area under the precision-recall
151 curve 0.52 to 0.14 for Pearson correlation and 0.32 to 0.13 for GENIE3, against 0.08 for a random ranking),
152 because volume and gene copy number induce positive correlations between all genes. Normalising counts to
153 concentrations removes the volume effect but recovers little (0.18 and 0.18), since the dependence on copy number
154 and cell age remains; regressing expression on the true cell-cycle covariates, which the simulation provides and
155 which cell-cycle scoring approximates in real data, recovers most of the loss (0.46 and 0.30). Sequencing at 15%
156 capture efficiency lowers accuracy further (0.11 raw and 0.24 library-size-normalised for Pearson correlation;
157 0.07 and 0.08 for GENIE3; Fig. 5b). Imputation with kNN-smoothing and MAGIC before inference changes
158 the area under the precision-recall curve by +0.05 to +0.21 (Pearson correlation) (Fig. 5c), consistent with the
159 original benchmark²⁵. Knocking down the five most connected regulators of a 20-gene network produces true
160 fold changes that a no-change baseline and the propagation of observational correlations predict with coefficients
161 of determination of -0.16 and -0.23 (Fig. 5d), illustrating how causal, out-of-distribution ground truth can be
162 generated for the evaluation of perturbation-prediction models^{13,14}.

163 **Inference recovers parameters and exposes the cost of ignoring division**

164 On a snapshot of 2,000 non-dividing cells, ABC-SMC with log-uniform priors over three decades and the
165 Wasserstein distance between simulated and observed counts²⁹ concentrates on the true telegraph parameters and
166 agrees with the exact maximum-likelihood estimate (posterior medians (0.31, 0.60, 20.9) for truth 0.3, 0.6, 20; Fig.
167 6a), the broadest posterior falling on the promoter inactivation rate, which a single snapshot constrains least. On
168 a snapshot of a growing and dividing population with volume-scaled transcription and mid-cycle replication, the
169 naive telegraph fit is biased ($k_{\text{on}} = 0.47$, $k_{\text{off}} = 1.95$, $k_{\text{tx}} = 71.5$) because volume and copy number inflate
170 counts and their dispersion, whereas ABC with the population simulator as forward model recovers the truth
171 ((0.30, 0.68, 20.8); Fig. 6b). Drug parameters follow from population data in the same way: from a kill curve
172 and the concordance of sister fates, the posterior over the maximal death rate and the protection threshold is
173 a ridge (weighted correlation of the two log parameters -0.85) that contains the truth (Fig. 6c), and posterior
174 predictive kill curves simulated at the posterior median reproduce the observed one (Fig. 6d). The two summaries
175 constrain the product of death rate and unprotected fraction, so separating the parameters requires measuring
176 the protective protein: an experimental-design question the forward model answers before an experiment is run.

177 **Calibration to time-lapse data predicts held-out conditions**

178 A simulator is useful for treatment questions only if it can be fitted to the data a laboratory produces and then
179 predict what it was not fitted to. We therefore calibrated the resistance-gene model to time-lapse measurements

in which U2OS cells were tracked for three days under 7, 10 and 13 μM cisplatin and every cell present at drug addition was scored as dying, dividing or surviving without dividing⁶ (Supplementary Table 4). Six parameters, the two promoter switching rates, the maximal death rate, the EC_{50} and Hill coefficient of death and the IC_{50} of growth arrest, were fitted to the fate fractions at 7 and 13 μM by simulated moments with common random numbers, under a weak prior keeping the memory of the resistant state near the two to three generations that the same study inferred from lineage correlations (Methods). The calibrated model (memory 2.8 generations, resistant fraction 15.2%, EC_{50} 16.7 μM ; Fig. 7f) reproduces the training fractions with a root-mean-square error of 0.071 and predicts the held-out 10 μM condition with an error of 0.093 (observed died, divided and survived fractions 0.28, 0.31 and 0.41; predicted 0.39, 0.20 and 0.41; Fig. 7a).

Properties that were not fitted can then be compared with the data. Death fates are correlated between sisters (0.09), first cousins (0.06) and second cousins (0.05), and the correlation decays to 0.02 for third cousins and 0.00 for unrelated cells (Fig. 7c), the pattern reported for HCT116 lineages. The mean time to death changes by 8% between 7 and 13 μM while the death fraction rises from 0.13 to 0.59 (Fig. 7d), the dose-invariance of single-cell timing that motivated the original study, and it arises here because dose changes the fraction of unprotected cells rather than their kinetics. The predicted kill curves (Fig. 7b) are 1.5-fold steeper at 13 μM in their first two days than in their second half, but the calibrated memory of 2.8 generations is too short to produce the plateau that HCT116 cells reach after about 100 h: in the model, cells leave the resistant state within days and are then killed. Reproducing the plateau requires a longer memory or a drug-induced stabilisation of the resistant state, which is the kind of conclusion a calibrated forward model can draw and a lineage-tracing experiment can test. The calibration landscape (Fig. 7e) shows how the fitted death rate and EC_{50} are constrained by the fate fractions alone; the protocol applies to any observable a laboratory records.

Schedule optimisation reproduces and explains clinical outcomes

With a calibrated model the framework becomes a schedule testbed. We first asked which schedule best delays regrowth of a melanoma-like population with a rare, heritable pre-resistant state² treated at a fractional-killing dose, comparing continuous dosing, the intermittent schedule of the SWOG S1320 trial, in which patients received a continuous 8-week lead-in and were then randomised to continuous dosing or to cycles of 3 weeks off and 5 weeks on³⁰, and adaptive therapy, in which the drug is withdrawn when the population has halved and resumed when it regrows to its initial size³¹. Because adaptive therapy deliberately lets the tumour regrow, schedules are compared by the time after randomisation at which the population exceeds 120% of its pre-treatment size (loss of control), with the RECIST-like time to progression from the nadir reported alongside (Methods).

The answer depends on two properties of the resistant cells (Fig. 8a–d). When the resistant state is fully protective and carries no fitness cost, as lineage data indicate for pre-existing tolerance⁶, resistant cells grow at the same rate with or without drug, holidays only let sensitive cells regrow and be killed again, and the three schedules lose control after 23, 26 and 27 weeks, the intermittent schedule delivering 65% of the cumulative dose. When resistant cells pay a 50% fitness cost in the absence of drug, the drug addiction that made intermittent dosing superior in patient-derived xenografts³², holidays let sensitive cells outcompete them: none of the three schedules loses control within the 52-week follow-up (52+, 52+ and 52+ weeks, censored), and adaptive therapy achieves this with 36% of the cumulative dose of continuous dosing. When resistant cells are only partly protected and grow at half speed under drug, holidays release them: continuous dosing keeps control for 52+ weeks against 49 for the intermittent schedule and 43 for adaptive therapy, the ranking observed in the trial, where continuous dosing gave a median progression-free survival of 9.0 months against 5.5 months for intermittent dosing. The optimiser makes the dependence explicit (Fig. 8e): over intermittent schedules of any period and duty cycle after

the lead-in, the best schedule is a period of 2.2 weeks with 39% of the time on drug (27 weeks to loss of control) without a fitness cost, a period of 1.0 weeks with 47% of the time on drug (52 weeks to loss of control) with a fitness cost and a period of 1.0 weeks with 100% of the time on drug (52 weeks to loss of control) with partial protection. The model therefore reproduces both the clinical and the preclinical outcome and attributes their difference to measurable properties of the resistant cells, their growth under drug and their fitness without it, which the framework can estimate from growth and lineage data before a schedule is chosen.

The second case study concerns glioblastoma, where the RTOG 0525 trial compared the standard temozolomide regimen (150–200 mg m⁻² on days 1–5 of 28-day cycles) with a dose-dense regimen (75–100 mg m⁻² on days 1–21) that delivers about twice the cumulative dose and was designed to deplete MGMT, and found no difference in survival for either MGMT-methylated or unmethylated tumours³³. We simulated both regimens with one-compartment pharmacokinetics (elimination half-life 2.1 h; ref.³⁴) on populations in which 1% (methylated) or 30% (unmethylated) of cells express MGMT, with MGMT protein protecting cells from the drug²⁷ and either stable or consumed by it, as a suicide enzyme is (Fig. 8f,g). With stable MGMT, the depth of response over six cycles is 0.20 log₁₀ units for the standard regimen and 1.00 for the dose-dense regimen in the methylated population (0.08 and 0.10 in the unmethylated one), and a 21-day regimen at the standard cumulative dose achieves no measurable kill (0.00 log₁₀ units): the death hazard rises steeply with concentration, so spreading the same dose over more days loses efficacy, and the advantage of the dose-dense arm in the model is entirely its larger cumulative dose. When the drug consumes MGMT, prolonged exposure depletes the protection of expressing cells: in the methylated population the dose-dense regimen deepens its response to 1.24 log₁₀ units against 0.21 for the standard regimen, and in the unmethylated population, where 30% of cells start protected, both regimens improve (final size 1.7 and 1.4 times the initial size, against 4.9 and 3.6 with stable MGMT) but neither controls the population, so the difference between the arms stays small (0.14 against 0.17 log₁₀ units), as in the trial. At the standard cumulative dose the number of dosing days per cycle that minimises the final population is 9 (methylated) and 9 (unmethylated) with stable MGMT, and 9 and 5 when MGMT is consumed (Fig. 8h). These simulations are not fits to the trial, whose end points depend on much that the model omits, dose-limiting toxicity among it, but they turn the rationale of dose-dense dosing into a quantitative prediction and identify the measurement that decides between the regimens: how fast the drug consumes MGMT in the tumour.

Discussion

Biomodelling.jl 2.0 fills a gap in the single-cell toolkit: a simulator in which expression memory, cell-size effects, cell-cycle dependence and drug-tolerant persistence follow from explicit biophysics instead of being assumptions of a generative model. That difference has practical consequences. As a hypothesis engine for non-genetic drug tolerance, the framework answers which schedule, which combination or which memory-modulating pretreatment minimises the long-term fitness of a treated population, with cell-level rules that can be matched to time-lapse and lineage-tracing data. As a ground-truth generator, it produces data with known memory genes, known networks, known perturbation effects and known lineage trees, which is what MemorySeq-style scores, lineage-aware models and virtual-cell benchmarks need in order to be evaluated fairly, and what existing simulators cannot supply. As a likelihood-free inference engine, it estimates mechanistic parameters from data that are now routinely collected, while quantifying the bias of simpler models.

The calibration and schedule studies set out a workflow for in silico single-cell pharmacology: declare the training and validation observables before fitting, fit by simulated moments or approximate Bayesian computation with common random numbers, validate on conditions and statistics that were not fitted, and only then compare regimens, reporting outcomes that map onto clinical end points. The case studies show why mechanism matters.

264 The same question, continuous or intermittent dosing, has opposite answers depending on whether resistant cells
 265 pay a fitness cost without drug, and a trial result and a xenograft result that appeared to contradict each other
 266 are both reproduced once that property is set. Because the framework simulates the lineage-resolved data from
 267 which such properties are measured, it can also say which experiment would settle the question before a trial is
 268 designed.

269 Several limitations delimit this release. Cells are well-mixed compartments without spatial structure or cell-cell
 270 signalling; the pharmacology is a one-compartment model; growth is exponential with phenomenological size
 271 control; and the inference layer, although general, relies on user-chosen summary statistics, so posterior calibration
 272 should be checked with predictive simulations. The persister case study is deliberately minimal, a single resistance
 273 gene with slow switching, and its calibration used fate fractions transcribed from a published study rather than raw
 274 tracks. The clinical case studies use literature values for cell-cycle times, resistant fractions and pharmacokinetics
 275 rather than patient-specific measurements, so their conclusions concern mechanisms and rankings, not absolute
 276 survival times. A bridge to the Catalyst.jl ecosystem is not part of this release. The benchmark results should be
 277 read as demonstrations of what the framework makes possible rather than as verdicts on the methods compared,
 278 which were run with default settings on small networks.

279 Spatial coupling through neighbourhood-dependent rates, cell-cell communication, chromatin states with their
 280 own inheritance rules and neural simulation-based inference are natural extensions of the existing forward model.
 281 Because the framework is deterministic given a seed, fast enough to simulate tens of thousands of cells in minutes,
 282 and writes AnnData files, it can be dropped into single-cell analysis pipelines as a source of mechanistically
 283 grounded synthetic data, and into virtual-cell training and evaluation as a source of causal, out-of-distribution
 284 contexts.

285 **Methods**

286 **Reaction model and propensities**

287 A model is a set of species and reaction channels. Species are molecules (partitioned at division) or promoter
 288 states (inherited). For a reaction with reactant counts x_i , stoichiometric coefficients ν_i , rate constant k and
 289 volume exponent e , the mass-action propensity is

$$a = k V^e \prod_i \binom{x_i}{\nu_i},$$

290 where by default $e = 1 - \sum_i \nu_i$, so that zero-order production scales with volume, first-order reactions are volume
 291 independent and bimolecular reactions scale as $1/V$; this keeps concentrations invariant under growth. Regulated
 292 reactions multiply this by $b + (1 - b)F(\mathbf{c})$, where b is a basal fraction and F combines Hill terms of the regulator
 293 concentrations $c_j = x_j/V$, $h_j = c_j^{n_j}/(K_j^{n_j} + c_j^{n_j})$ for activators and $1 - h_j$ for inhibitors, either as a product
 294 (AND logic) or as $1 - \prod_j (1 - h_j)$ (OR logic). Promoter switching is represented by explicit promoter species (for
 295 instance $G_{\text{off}} \rightleftharpoons G_{\text{on}}$) grouped into promoter groups. Custom propensities can be supplied as functions of counts,
 296 volume, parameters and time.

297 **Stochastic simulation kernels**

298 Within one update step the cell volume and gene copy number are held fixed and the reaction network is advanced
 299 by one of four kernels: the direct method³⁵ with propensities updated through a species-reaction dependency

graph; fixed-step Poisson tau-leaping³⁶; a hybrid scheme in which a leap that would drive any species negative is rejected and the interval is instead simulated exactly (this is the exact version of the “switching” scheme of version 1); and adaptive tau-leaping with the step-size selection of³⁷ and the critical-reaction partition of³⁸, falling back to a fixed number of exact events when the proposed leap is shorter than $10/a_0$. Every kernel takes an explicit random number generator.

Cells, growth, replication and division

Each cell carries its volume V , age, generation, lineage identifiers, gene copy state, growth rate, molecule counts, a cell-specific parameter vector and its own random number generator (seeded from the parent’s generator, which makes results independent of the number of threads). Volume grows exponentially, $\dot{V} = \lambda V$, with an optional log-normal spread of λ between cells. Division is triggered by a sizer ($V \geq V_{\text{div}}$), an adder ($V - V_{\text{birth}} \geq \Delta$) or a timer (age $\geq T$), each with a per-cell noise factor drawn at birth. At division the first daughter inherits a volume fraction $f \sim \mathcal{N}(0.5, \sigma^2)$ (clipped to $[0.05, 0.95]$) and each molecule is assigned to it independently with probability f (binomial partitioning); a beta-binomial option models over-dispersed partitioning. Promoter groups are inherited by both daughters when unreplicated and split one copy per daughter, in random assignment, when replicated. Gene replication, if enabled, doubles the promoter counts (and the propensity of reactions flagged as copy-number dependent) once the cell has completed a set fraction of its cycle, and is reset at division. Populations can be started from per-founder states and volumes (for instance a snapshot of a burnt-in constant-size population). Population control is either constant size (every second daughter replaces a uniformly chosen cell, the behaviour of version 1), free growth (all daughters kept, with uniform subsampling above a cap and book-keeping of the true size) or logistic (free growth with an additional hazard $\lambda N/K$).

Drug and perturbation layer

A perturbation combines a dose schedule $d(t)$ (constant, pulsed, piecewise, one-compartment bolus pharmacokinetics or an arbitrary function) with effects evaluated for every cell at every step: a death hazard

$$h = h_{\max} \frac{d^m}{\text{EC}_{50}^m + d^m} \cdot \frac{K^q}{K^q + c_P^q},$$

where c_P is the concentration of a protective species (survival probability over a step is $e^{-h \Delta t}$); growth inhibition by $1/(1 + (d/\text{IC}_{50})^m)$; a dose-independent fitness cost of a cell state, $1 - c_{\max} c_P^q/(K^q + c_P^q)$, on the growth rate; multiplicative modulation of any parameter by a function of the dose; and genetic perturbations (knockdown, knockout, over-expression) of any parameter in a random fraction of cells, the flag being inherited by daughters.

Lineage statistics

The lineage table records, for every cell, its parent, founder (clone), generation, birth and end times, fate (divided, died, removed, alive) and its volume and state at birth and at the end of its life. From it we compute Pearson correlations of expression at division between mothers and daughters, sisters and cousins; the lineage autocorrelation r_g between a cell and its ancestor g generations earlier, and the memory timescale τ from $r_g = e^{-g/\tau}$; single-lineage traces by following the ancestral line of a surviving cell; the coefficient of variation squared of concentration across the population versus along single lineages; Luria-Delbrück fluctuation tests (variance of a phenotype across clones grown from single cells compared with the binomial expectation); and MemorySeq-style clonal scores³⁹, the variance of clone-mean expression divided by its permutation-null expectation.

Observation models

True counts are converted to observations by a scRNA-seq model in which each cell captures every molecule independently with a Beta-distributed efficiency (mean and coefficient of variation set by the protocol), optional multinomial down-sampling to a Poisson-distributed depth, optional additional dropout and batch scaling; an smFISH model with a fixed detection efficiency; and a time-lapse model that samples a species along lineages at a fixed interval with additive or multiplicative measurement noise. Because counts scale with volume, library size correlates with cell size, reproducing the transcriptome-size confound of real data. Outputs are written as CSV or as AnnData-compatible HDF5 files.

Inference

Snapshot counts of a gene in non-dividing cells follow the Beta-Poisson law of the telegraph model²⁶, which we evaluate through Kummer’s confluent hypergeometric function with rescaled series summation, and maximise by a grid-initialised Nelder-Mead search in log-parameter space. For any model, including populations, we implement sequential Monte Carlo approximate Bayesian computation⁴⁰ with an adaptive tolerance schedule (the α quantile of the previous generation’s distances), Gaussian perturbation kernels of variance twice the weighted population variance, importance weights against the prior, and parallel evaluation of the user-supplied distance function; for count distributions we use the Wasserstein-1 distance between simulated and observed samples²⁹.

Calibration to laboratory data and validation strategy

Reference values were transcribed from the text and tables of the cited articles into `paper/data/` with their provenance (Supplementary Table 4). For the cisplatin case study the model is a resistance gene with slow promoter switching whose protein product protects against a dose-dependent death hazard and against a dose-dependent growth arrest (Methods above), with a 24-h cell cycle. Its six free parameters (activation and inactivation rates, maximal death rate, EC_{50} and Hill coefficient of death, IC_{50} of growth arrest) were fitted by the method of simulated moments: the root-mean-square difference between simulated and observed fractions of cells that died, divided or survived without dividing during 72 h of drug, for the training concentrations (7 and 13 μ M), plus a weak penalty keeping the memory of the resistant state near the 2-3 generations inferred from lineage correlations, was minimised with the bounded optimiser described below using common random numbers (the same seeds for every candidate). Validation used only quantities not fitted: the fate fractions at the held-out concentration (10 μ M), the biphasic kill curve and its plateau, the extent of lineage correlations in fate (sisters, first, second and third cousins) and the invariance of single-cell division and death times across concentrations. The protocol is deliberately generic: any observable computable from a `PopulationResult` (kill curves, fate fractions, lineage correlations, snapshot distributions) can enter the distance, and the split between training and validation conditions is declared before fitting.

Clinical regimens, treatment outcomes and schedule optimisation

Regimens are built from the schedule primitives of the perturbation layer: `PulsedDose` with a finite number of cycles, `PiecewiseDose` for arbitrary calendars (the S1320 schedule), `daily_boluses` and `cycle_days` for daily oral doses with one-compartment pharmacokinetics, and `AdaptiveDose`, a feedback schedule that withdraws the drug when the population falls below a fraction of its initial size and resumes it when the population regrows above another fraction, evaluated with the current population size at every step³¹. Outcomes computed from a simulation are the net growth rate, the depth of response ($\log_{10}$ of the ratio of the initial to the smallest population), the time to progression (first time the population exceeds a threshold, 1.2, times its running minimum, the analogue

of RECIST progression from the nadir; censored at the end of the simulation), the extinction probability over replicates and the cumulative dose actually applied. `optimize_schedule` minimises any such outcome over a box of schedule parameters: a log-spaced grid (or a Latin hypercube when the grid is large) is evaluated with the same random seeds for every candidate, and the best point is refined by Nelder-Mead in a bounded transform of the box; constraints such as a fixed cumulative dose enter as penalties or as the parameterisation (for instance a per-day dose equal to the budget divided by the number of dosing days). The case studies are run at the scale of a tumour rather than a culture: the cell cycle of the model is the net doubling time of the tumour (4 weeks for melanoma, 40 days for glioblastoma), gene-expression rates are scaled to the hour (mRNA half-life 7 h, protein half-life 35 h) and founders are drawn from the stationary distribution of promoter states. The melanoma case uses a pre-resistant fraction of 1:200 with a memory of five generations^{2,39} and drug effects at the fractional-killing dose: sensitive cells arrest and die with a half-life of two weeks, cells in the resistant state are protected, and the drug stabilises the resistant state ten-fold (the drug-induced reprogramming of²). Three variants of the resistant state are compared: full protection without cost; a 50% fitness cost in the absence of drug³²; and partial protection, in which resistant cells grow at half speed under drug. Progression is scored in two ways: from the nadir, as in RECIST (a 20% increase in diameter, taken as a 1.73-fold increase in cell number, after the 8-week lead-in), and as loss of control, the time after randomisation at which the population exceeds 120% of its pre-treatment size, the end point of adaptive-therapy studies, which is the one used to compare schedules that deliberately let the tumour regrow. The temozolomide case uses an elimination half-life of 2.1 h³⁴, the regimens of RTOG 0525³³ (200 mg/m² on days 1-5 or 100 mg/m² on days 1-21 of 28-day cycles, six cycles), MGMT-expressing fractions of 1% and 30% for methylated and unmethylated promoters, and MGMT either stable or consumed by the drug (its degradation rate increased 20-fold at the standard peak concentration), the suicide-enzyme behaviour that motivated dose-dense schedules.

Random regulatory networks

Random networks are built with a chosen number of activating and inhibiting edges under Erdős-Rényi or preferential-attachment (scale-free) sampling; regulation acts on the transcription rate of constitutive genes or on the promoter activation rate of telegraph genes through AND-combined Hill functions; per-gene rate constants can be drawn log-uniformly from ranges.

Validation protocols

Exactness was tested by Kolmogorov-Smirnov distances between simulated stationary distributions and the analytic laws (Poisson for birth-death, Beta-Poisson for the telegraph model, negative binomial for bursty protein production^{41,42}), by agreement between kernels, and by a two-sample comparison against `JumpProcesses.jl`⁴³ on the same model. Population-level properties were compared with the analytic expectations for cell-size scaling and concentration homeostasis²³, partitioning noise⁴⁴, lineage versus population statistics¹¹ and cell-cycle-dependent burst parameters¹⁰. Persister simulations were compared with the quantitative signatures reported by^{6,28} and⁹, and with the phenotypic-selection model of²⁷. All scripts, seeds and outputs are in the `paper/` directory of the repository.

Statistics and reproducibility

No statistical method was used to predetermine sample size, and no experimental data were generated in this study; sample sizes are the numbers of simulated cells, founders or seeds stated in each figure legend and in Supplementary Table 2. No data were excluded. Randomisation and blinding do not apply: every result is

416 a deterministic function of an explicit random seed, and all seeds are fixed in the scripts that generate the
417 figures. Replicates are independent simulations differing only in seed; where a mean and a standard deviation
418 are reported, the number of seeds is given in the legend. Kolmogorov-Smirnov distances are compared with the
419 99% critical value for the stated sample size. Distributional agreement between kernels and with an independent
420 implementation was assessed by two-sample Kolmogorov-Smirnov statistics. Correlations are Pearson correlations
421 unless stated otherwise; correlations of binary fates between related cells are phi coefficients. Posterior summaries
422 are medians, because the posteriors are asymmetric on the log scale. Multithreaded runs give results identical to
423 single-threaded runs, because each cell carries its own random number generator seeded from its parent.

424 **Data availability**

425 The laboratory and clinical reference values used for calibration and validation were transcribed from the published
426 articles cited in Supplementary Table 4 and are provided, with their provenance, in the **paper/data/** directory
427 of the software repository (<https://github.com/ayoublasri/Biomodelling.jl>). All other data in this Article are
428 simulated and are regenerated from fixed seeds by the scripts in the **paper/** directory; the generated tables are
429 archived with the software release.

430 **Code availability**

431 Biomodelling.jl 2.0 is open source under the MIT licence at <https://github.com/ayoublasri/Biomodelling.jl>, with
432 documentation, tutorials and a test suite that runs under continuous integration. Every figure is regenerated by
433 `julia --project=paper paper/run_all.jl` followed by `python paper/plot_all.py`.

434 **Software**

435 Biomodelling.jl 2.0 is written in Julia (≥ 1.10) and depends on Distributions.jl, PoissonRandom.jl, QuadGK.jl,
436 Optim.jl, SpecialFunctions.jl and StatsBase.jl; HDF5.jl is an optional extension. Benchmarks used absolute
437 Pearson and Spearman correlations and scikit-learn random forests re-implementing GENIE3⁴⁵, MAGIC⁴⁶ and
438 kNN-smoothing⁴⁷. The test suite (Aqua quality checks and statistical tests) runs under GitHub Actions on Julia
439 1.10 and the latest release.

Figure legends

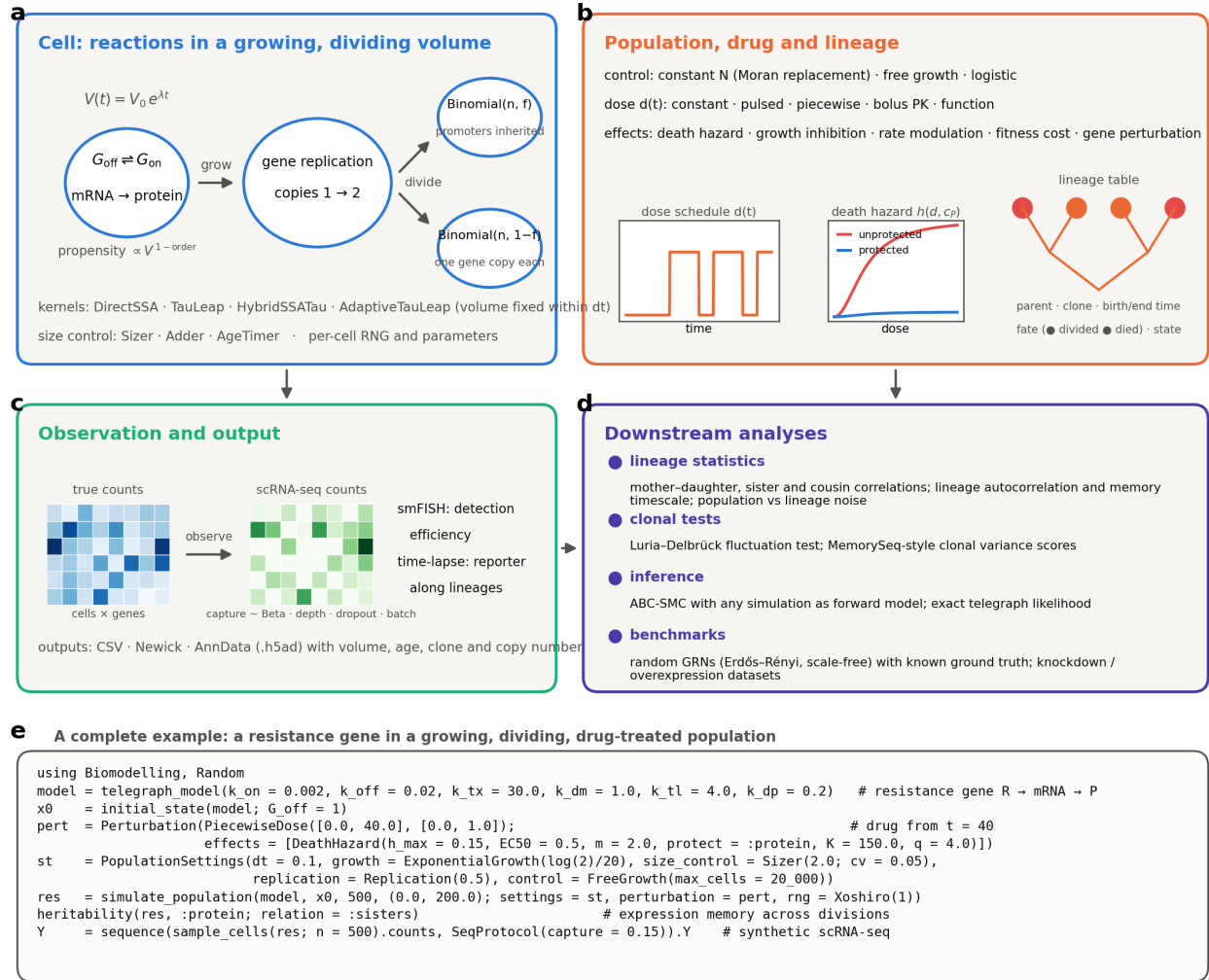


Fig. 1 | The Biomodelling.jl 2.0 framework. **a**, Each cell runs the stochastic kinetics of a reaction network (mass-action, Hill or custom propensities, explicit promoter states) at its current volume; volume grows exponentially, genes replicate at a set point of the cycle, and division follows a sizer, adder or timer rule with binomial partitioning of molecules and inheritance of promoter states. **b**, Population control (constant size, free growth or logistic), the perturbation layer (dose schedules, state-dependent death, growth inhibition, rate modulation, genetic perturbations) and the lineage table that records every cell's ancestry, fate and state at birth and division. **c**, Observation models (scRNA-seq with capture, depth, dropout and batch effects; smFISH; time-lapse reporters) and outputs (CSV, Newick, AnnData). **d**, Downstream uses: lineage statistics, fluctuation tests, clonal memory scores, likelihood-free inference and benchmark generation. **e**, A complete example in ten lines of code.

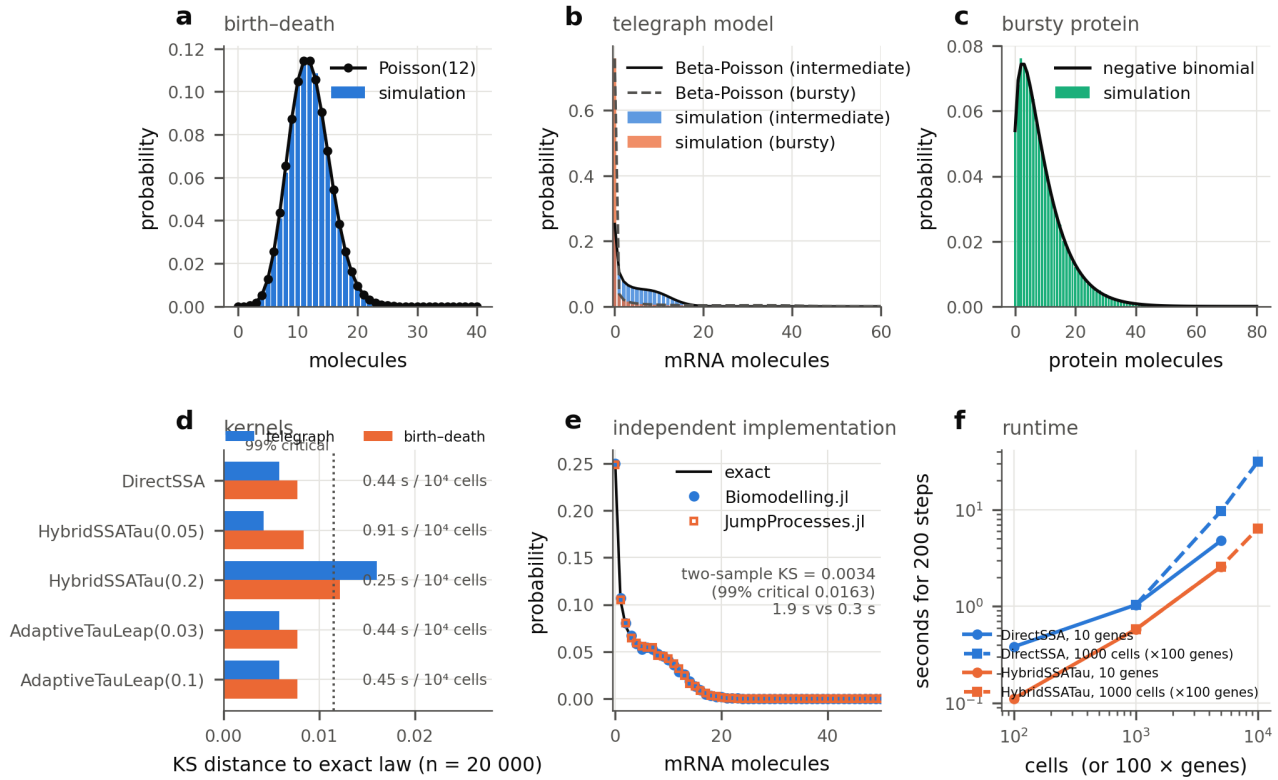
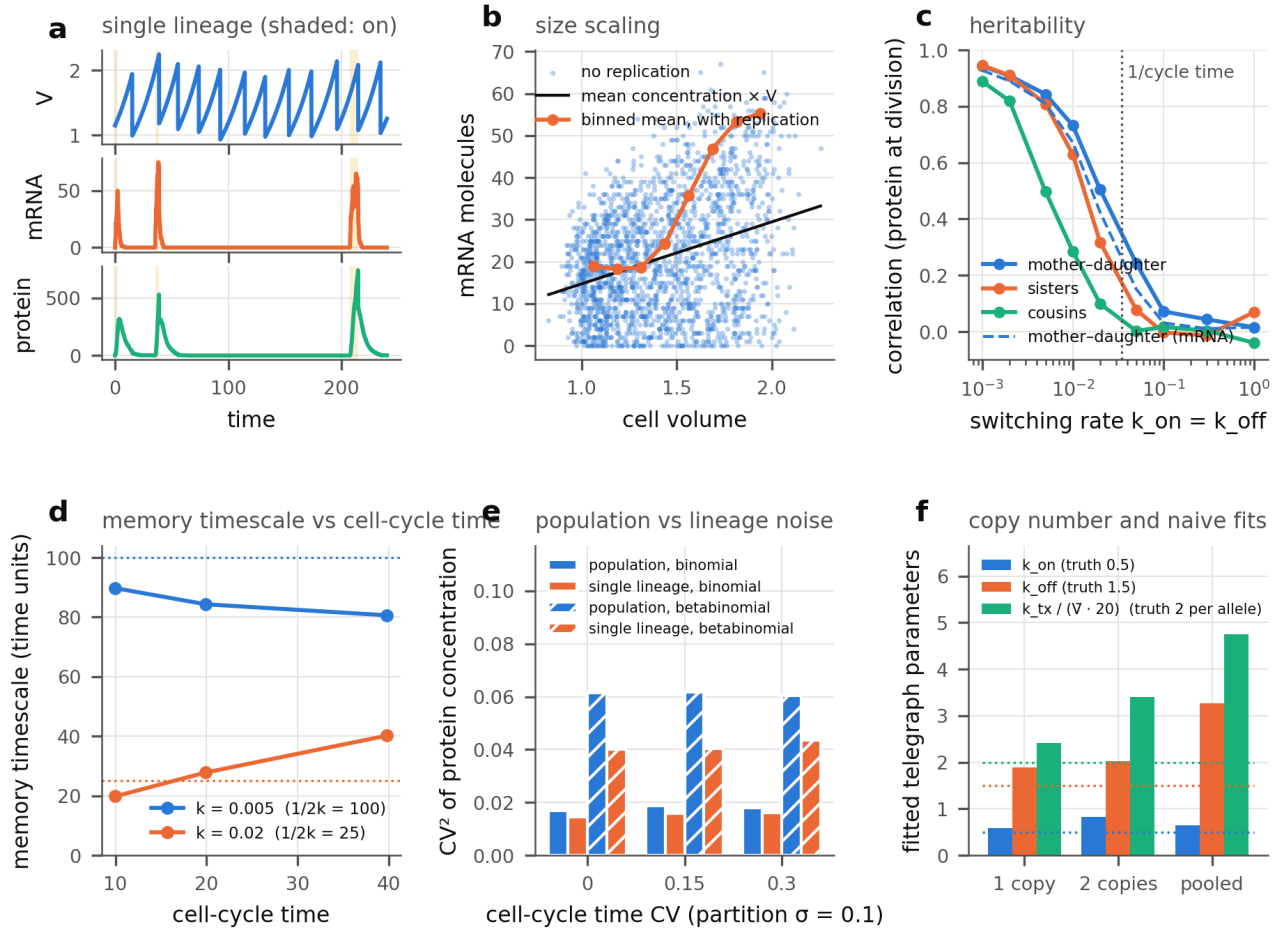


Fig. 2 | The engine is exact and agrees with an independent implementation. **a**, Stationary distribution of a birth-death process (bars, 50,000 simulated cells; line, Poisson law). **b**, Telegraph model in an intermediate and a bursty regime against the Beta-Poisson law. **c**, Bursty protein production against the negative binomial. **d**, Kolmogorov-Smirnov distance to the exact law for every kernel (n = 20,000; dotted line, 99% critical value) with wall-clock time per 10^4 cells. **e**, Telegraph model simulated with Biomodelling.jl and with JumpProcesses.jl on the same model; the two-sample Kolmogorov-Smirnov statistic lies below its 99% critical value. **f**, Runtime of population simulations against the number of cells (10 genes) and genes (1,000 cells) for the exact and hybrid kernels.



461

462 **Fig. 3 | Growth, division and partitioning generate memory, size scaling and cell-cycle effects.** **a**, A
 463 single lineage: volume, mRNA and protein of a slowly switching gene, with the promoter-on state shaded. **b**,
 464 mRNA count against cell volume for a fast-switching gene without replication (points; line, mean concentration $\times V$) and with gene replication at mid-cycle (binned means). **c**, Correlation of protein concentration at division
 466 between mothers and daughters, sisters and cousins as a function of the promoter switching rate; dotted line,
 467 inverse cell-cycle time. **d**, Memory timescale against the measured cell-cycle time for two switching rates; dotted
 468 lines, $1/(k_{\text{on}} + k_{\text{off}})$. **e**, Squared coefficient of variation of a stable protein's concentration across the population
 469 and along single lineages, for increasing partitioning noise and cell-cycle-time noise. **f**, Telegraph parameters fitted
 470 to cells before replication (one gene copy), after replication (two copies) and pooled; dotted lines, per-allele values.

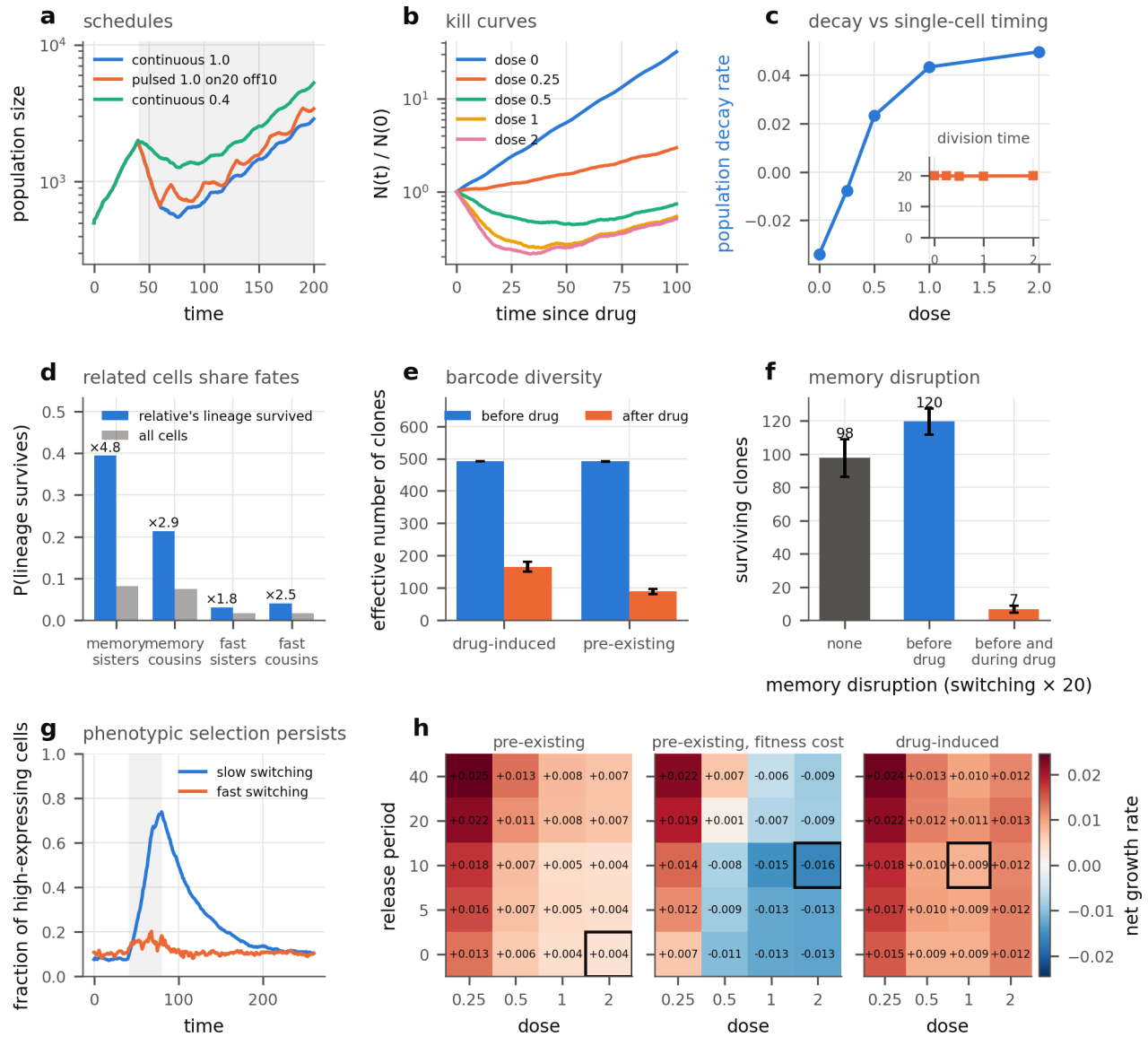
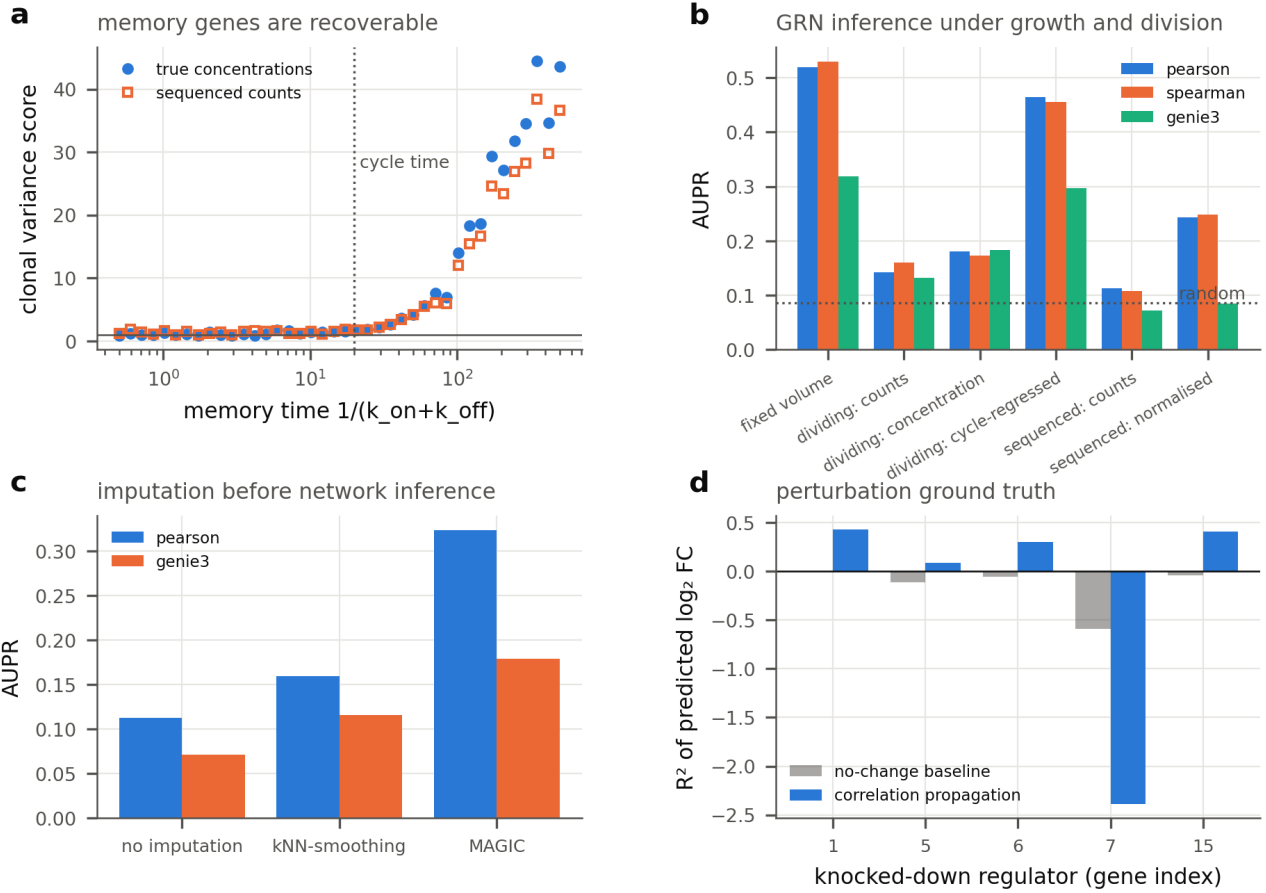


Fig. 4 | A single resistance gene with slow promoter switching reproduces persister biology. **a**, Population size under continuous, pulsed and reduced-dose schedules (shading, drug present). **b**, Kill curves at increasing dose. **c**, Net population decay rate over the first 30 time units of treatment against dose, with the mean division time of cells born under drug (inset), which is unchanged. **d**, Probability that a cell's lineage survives the treatment (at least one descendant alive at the end of the run) given that its sister's or cousin's lineage survived, against the marginal survival probability, for cells born in the 25 time units before the drug; memory gene and fast-switching control, with the fold enrichment above each bar. **e**, Effective number of clones (barcodes) before and after drug for pre-existing and drug-induced tolerance (mean $\pm$ s.d. of three seeds). **f**, Surviving clones after drug without memory disruption, with promoter switching accelerated 20-fold from two cell-cycle times before the drug until the drug arrives, and with the acceleration maintained during exposure (mean $\pm$ s.d. of five seeds). **g**, Fraction of high-expressing cells before, during and after a drug pulse for slow and fast switching. **h**, Net population growth rate over a 200-time-unit course as a function of release period (time units without drug between 20-time-unit exposures) and dose, for pre-existing tolerance without and with a fitness cost of the resistant state and for drug-induced tolerance; the box marks the most effective schedule.



486

487 **Fig. 5 | Ground-truth benchmarks.** **a**, Clonal variance score against the known memory time of 40
 488 independent genes, from true concentrations and from sequenced counts; dotted line, cell-cycle time. **b**, Area
 489 under the precision-recall curve of correlation (Pearson, Spearman) and GENIE3 network inference on the same
 490 30-gene network observed at fixed volume, in a dividing population (raw counts, concentrations, and concentrations
 491 regressed on the cell-cycle covariates copy number and age) and after sequencing (raw and library-size normalised
 492 counts); dotted line, random ranking. **c**, Effect of kNN-smoothing and MAGIC imputation before inference. **d**,
 493 Coefficient of determination of predicted $\log_2$ fold changes after knocking down the five most connected regulators,
 494 for a no-change baseline and for propagation of observational correlations.

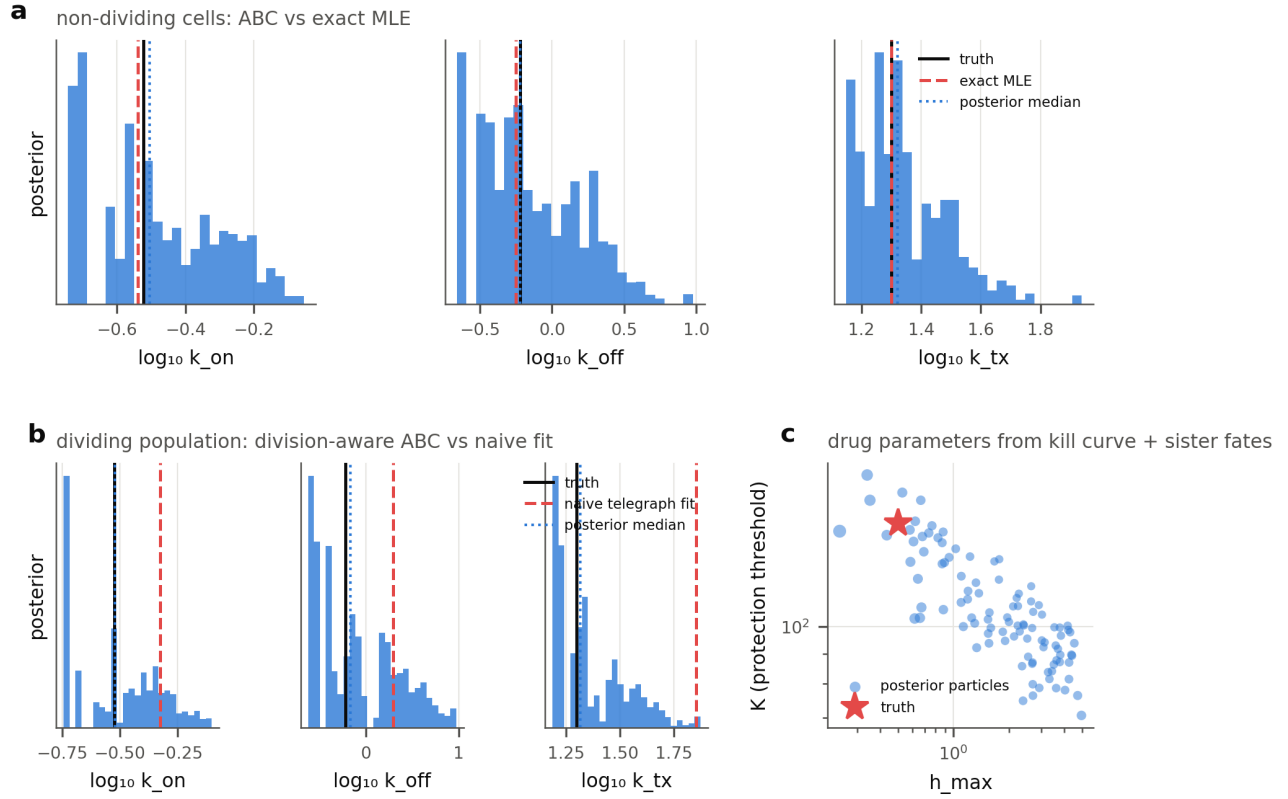


Fig. 6 | Inference with the forward model. **a**, Posterior distributions of telegraph parameters from ABC-SMC on a snapshot of non-dividing cells, with the truth and the exact maximum-likelihood estimate. **b**, The same for a snapshot of a growing and dividing population: the naive telegraph fit is biased while division-aware ABC recovers the truth. **c**, Posterior particles for the maximal death rate and the protection threshold inferred from a kill curve and sister fate concordance: the two parameters trade off along a ridge that contains the truth. **d**, Posterior predictive kill curves simulated at the posterior median against the observed curve.

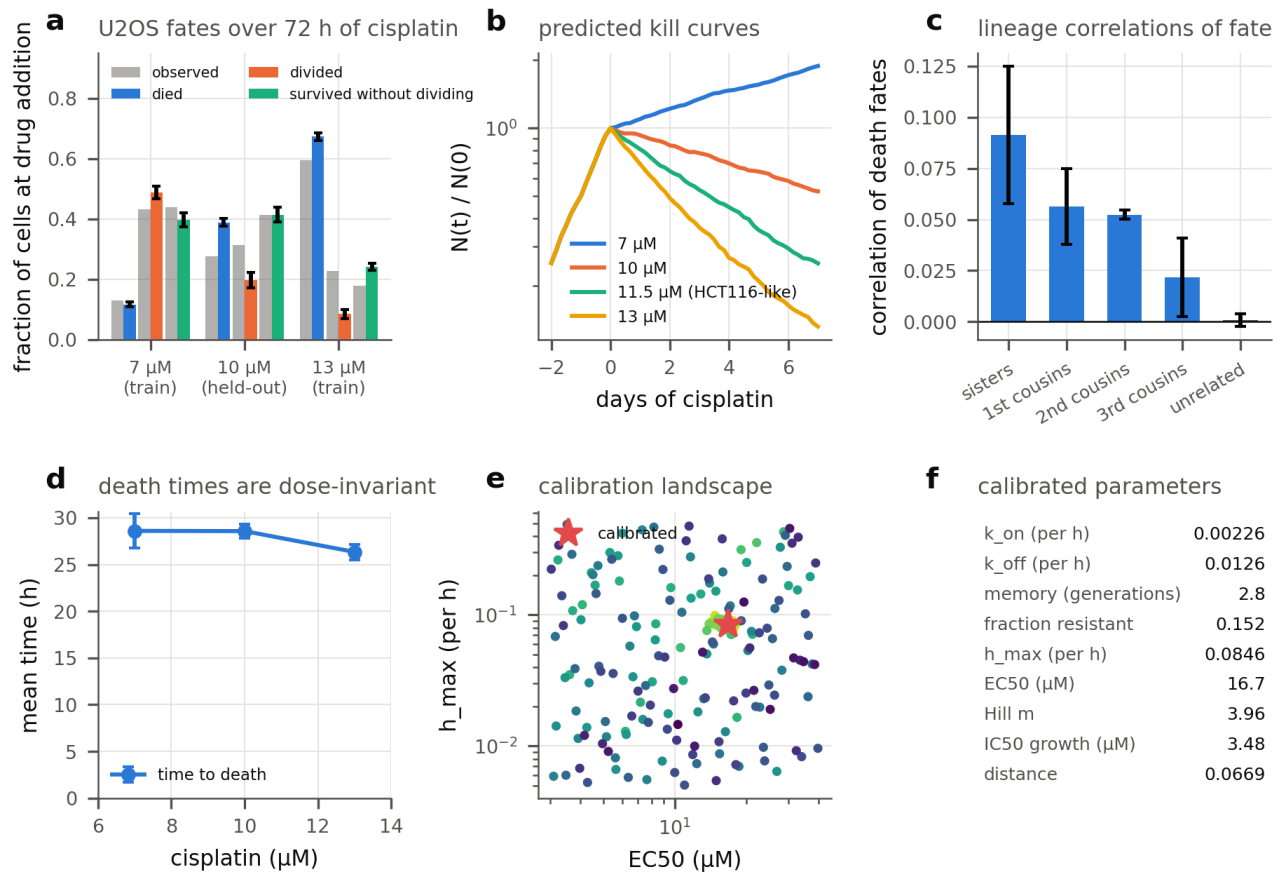
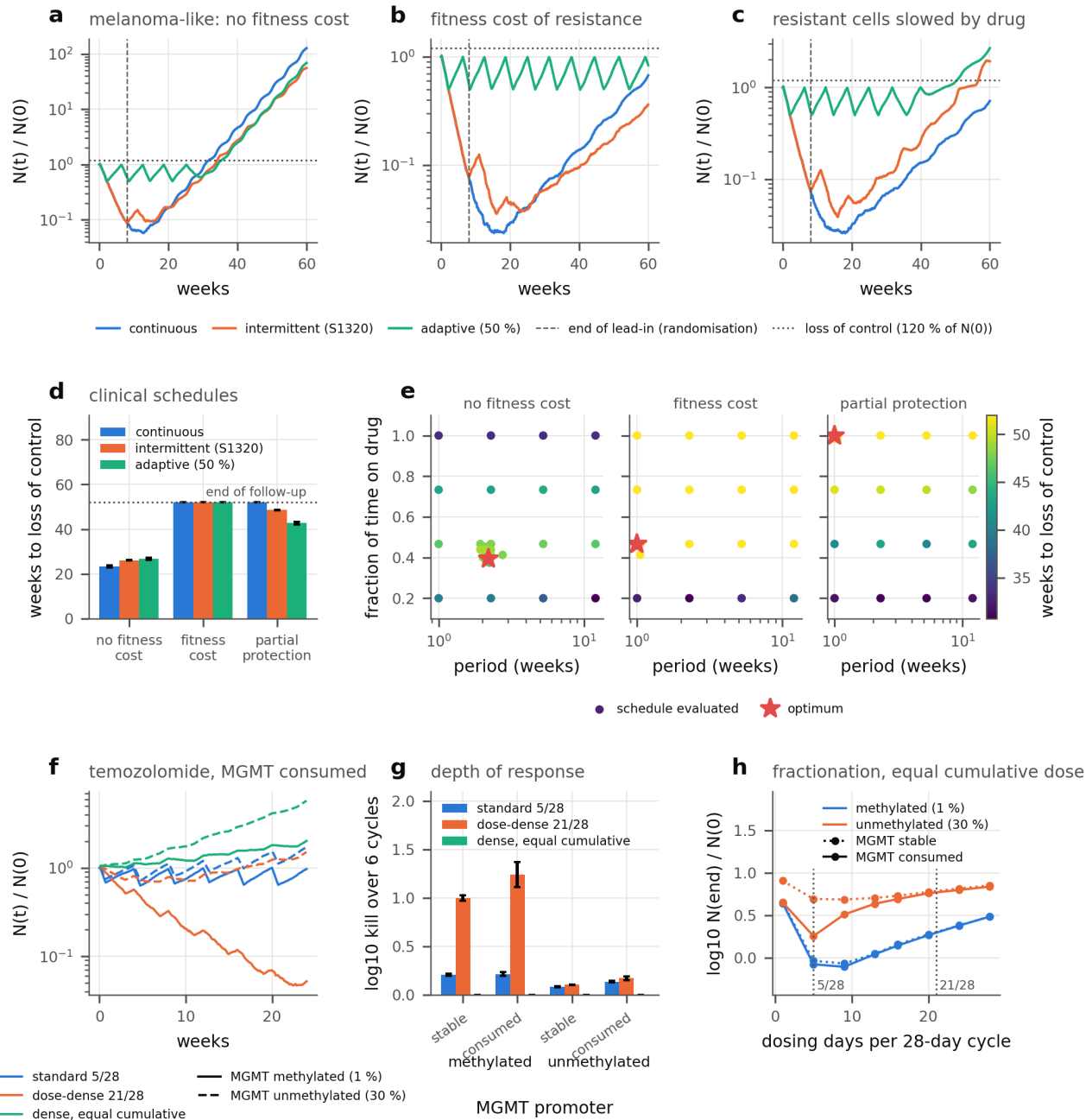


Fig. 7 | Calibration to laboratory time-lapse data and validation on held-out conditions. **a**, Fates over 72 h of cisplatin of U2OS cells present at drug addition, observed by time-lapse microscopy (grey; ref.⁶) and simulated by the calibrated model (colours; mean $\pm$ s.d. of four seeds); the model was calibrated on 7 and 13 μM and the 10 μM condition is predicted. **b**, Predicted population size at the three concentrations and at 11.5 μM , from two days before the drug to seven days of exposure. **c**, Correlation of death fates between sisters, first, second and third cousins present at drug addition, and between unrelated cells (mean $\pm$ s.d. of three seeds), against the observation that correlations extend to second but not third cousins. **d**, Mean time to death of cells present at drug addition, and mean division time where at least 20 cells born under drug divided, against concentration (mean $\pm$ s.d. of four seeds). **e**, Calibration landscape: every parameter set evaluated by the optimiser, coloured by its distance to the training data. **f**, Calibrated parameters.



513

514 **Fig. 8 | Dose and schedule optimisation against clinical regimens.** **a–c**, Population size, relative to its
515 pre-treatment value, under continuous dosing, the intermittent schedule of the SWOG S1320 trial (8-week lead-in,
516 dashed line, then 3 weeks off and 5 weeks on) and adaptive therapy (drug withdrawn below 50% of the initial size,
517 resumed at the initial size) for a melanoma-like population with a rare, heritable pre-resistant state, when the
518 resistant state carries no fitness cost (**a**), when resistant cells grow 50% slower in the absence of drug (**b**), and
519 when resistant cells are only partly protected and grow at half speed under drug (**c**); dotted line, 120% of the
520 initial size. **d**, Weeks after randomisation until the population exceeds 120% of its pre-treatment size (loss of
521 control; mean $\pm$ s.d. of four seeds, censored at 52 weeks). **e**, The same end point over intermittent schedules of
522 given period and fraction of time on drug after the lead-in, for every schedule evaluated by the optimiser (star,
523 best), for the three mechanisms. **f**, Population size under the standard (5 of 28 days) and dose-dense (21 of 28

days) temozolomide regimens of the RTOG 0525 trial and a 21-day regimen at the standard cumulative dose, with one-compartment pharmacokinetics (elimination half-life 2.1 h) and consumption of MGMT by the drug, for populations in which 1% (methylated promoter, solid) or 30% (unmethylated, dashed) of cells express MGMT. **g**, Depth of response over six cycles for the three regimens, the two populations and MGMT either stable or consumed by the drug (mean $\pm$ s.d. of two seeds). **h**, Final population size against the number of dosing days per 28-day cycle at the standard cumulative dose.

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Author contributions

A.L. conceived the study, defined its scope, the validation strategy and the case studies, supervised the work, verified all results and wrote the final version of the manuscript. The implementation of the software, the simulations, the analyses, the figures and the first draft of the manuscript were produced with Claude (Anthropic) under the author’s direction.

Competing interests

The author declares no competing interests.

Additional information

Supplementary information accompanies this Article. Correspondence and requests for materials should be addressed to A.L.

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